

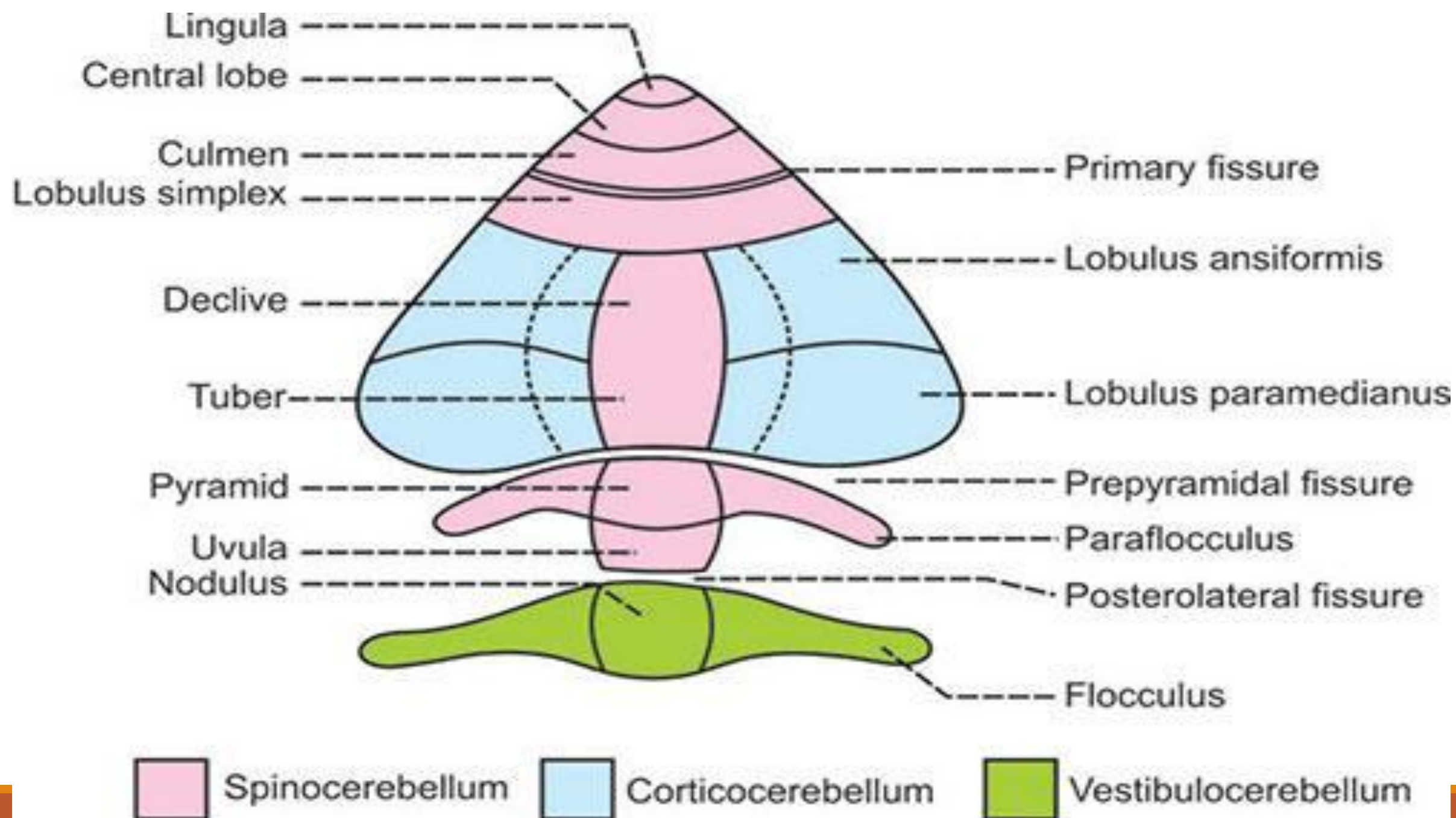
INPUT & OUTPUT CONNECTIONS OF CEREBELLUM

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LEARNING OBJECTIVES

By the end of this lecture, students should be able to:

- Identify the location & major external & internal structures of cerebellum.
- Describe anatomical divisions of cerebellum(lobes, hemispheres, vermis, cortex & nuclei)
- Explain input(afferent) & output(efferent) pathways in relation to functional divisions of cerebellum.
- Describe the major functions of cerebellum.
- Identify the effects of cerebellar damage in clinical practice.



VESTIBULOCEREBELLUM (ARCHICEREBELLUM)

- Vestibulocerebellum is the oldest part of cerebellum phylogenetically hence called as ***archicerebellum***.
- It includes flocculonodular lobe formed by ***nodulus*** of vermis & its lateral extensions ***flocculi***.
- Vestibulocerebellum maintains posture & equilibrium.

CONNECTIONS OF VESTIBULOCEREBELLUM

AFFERENT CONNECTIONS

Vestibulocerebellar tract

- ❑ Formed by fibers arising from vestibular nuclei situated in medulla & pons.
- ❑ It passes through the inferior cerebellar peduncle of same side & reaches cerebellar nuclei, nucleus globosus, nucleus emboliformis & nucleus fastigi.
- ❑ Fibers from these nuclei reach the vestibulocerebellum.
- ❑ Vestibular nuclei receive nerve fibers from the vestibular apparatus present in the inner ear through vestibular division of VIII cranial nerve(cochlear nerve).

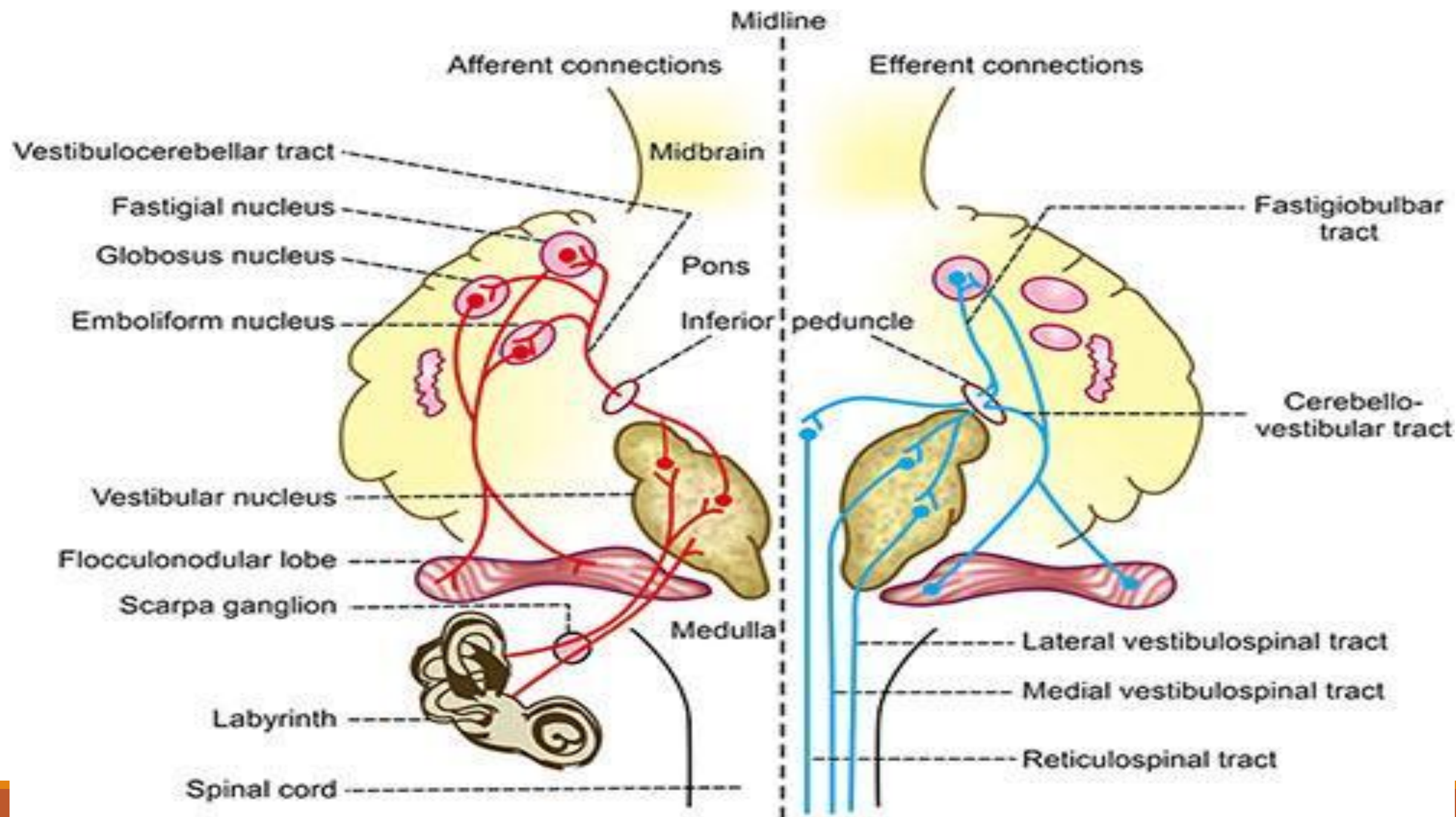
EFFERENT CONNECTIONS

Cerebellovestibular tract

- Fibers arise from flocculonodular lobe & run through inferior cerebellar peduncle of same side & end on the vestibular nuclei situated in the brainstem.
- Fibers from vestibular nuclei form medial & lateral vestibulospinal tracts & terminate on the medial group of alpha motor neurons in the spinal cord.

Fastigiobulbar tract

- Fibers arise from fastigial nucleus pass through inferior cerebellar peduncle of same side & end on vestibular nuclei & reticular formation in medulla oblongata.
- Fibers of vestibulospinal tract arising from vestibular nuclei terminate on alpha motor neurons in the spinal cord.
- Fibers of reticulospinal tract arise from reticular formation & terminate on gamma motor neurons in spinal cord.



FUNCTIONS OF VESTIBULOCEREBELLUM

- Vestibulocerebellum regulates *muscle tone, posture & equilibrium* by receiving impulses from the vestibular apparatus.
- Lesions of vestibulocerebellum result in reduced muscle tone(hypotonia) & failure to maintain posture & equilibrium.

SPINOCEREBELLUM (PALEOCEREBELLUM)

- Spinocerebellum is connected with the spinal cord & forms major receiving area of cerebellum for sensory inputs.
- It includes *medial portions of cerebellar hemisphere, paraflocculi* & parts of vermis i.e. *lingula, central lobe, culmen, lobulus simplex, declive, tuber, pyramid & uvula*.
- Spinocerebellum is associated with the maintenance of muscle tone & anticipatory adjustment of muscle contraction during movement.

CONNECTIONS OF SPINOCEREBELLUM

AFFERENT CONNECTIONS

Dorsal spinocerebellar tract

- Fibers of this tract arise from Clarke's column of cells in the dorsal gray horn of spinal cord & reaches spinocerebellum through inferior peduncle of same side.
- This tract conveys proprioceptive information regarding position & movements from limbs of same side.

Ventral spinocerebellar tract

- ❑ Fibers of this tract arise from marginal cells in the dorsal gray horn of spinal cord.
- ❑ After originating, fibers cross the midline & ascend in opposite side to reach spinocerebellum through superior cerebellar peduncle.
- ❑ This tract conveys proprioceptive signals to spinocerebellum regarding position & movements of opposite limbs.

Cuneocerebellar tract

- ❑ Fibers arise from accessory cuneate nucleus situated lateral to cuneate nucleus of medulla & reaches spinocerebellum through inferior cerebellar peduncle of same side.
- ❑ This tract send proprioceptive impulses to spinocerebellum from upper limb, upper trunk & neck

Olivocerebellar tract

- Olivocerebellar tract is formed by climbing fibers originating from inferior olivary nucleus of medulla.
- The fibers then cross the midline & reach spinocerebellum through inferior cerebellar peduncle of opposite side.
- Inferior olivary nucleus receive afferent fibers from brainstem nuclei of same side, spinal cord through spino-olivary tract of same side & cerebral cortex of opposite side.
- Olivocerebellar tract also gives collaterals to cerebellar nuclei especially to nucleus globosus & nucleus emboliformis.
- This tract sends proprioceptive impulses from the body & also conveys output signals from cerebral cortex to spinocerebellum.

Pontocerebellar tract

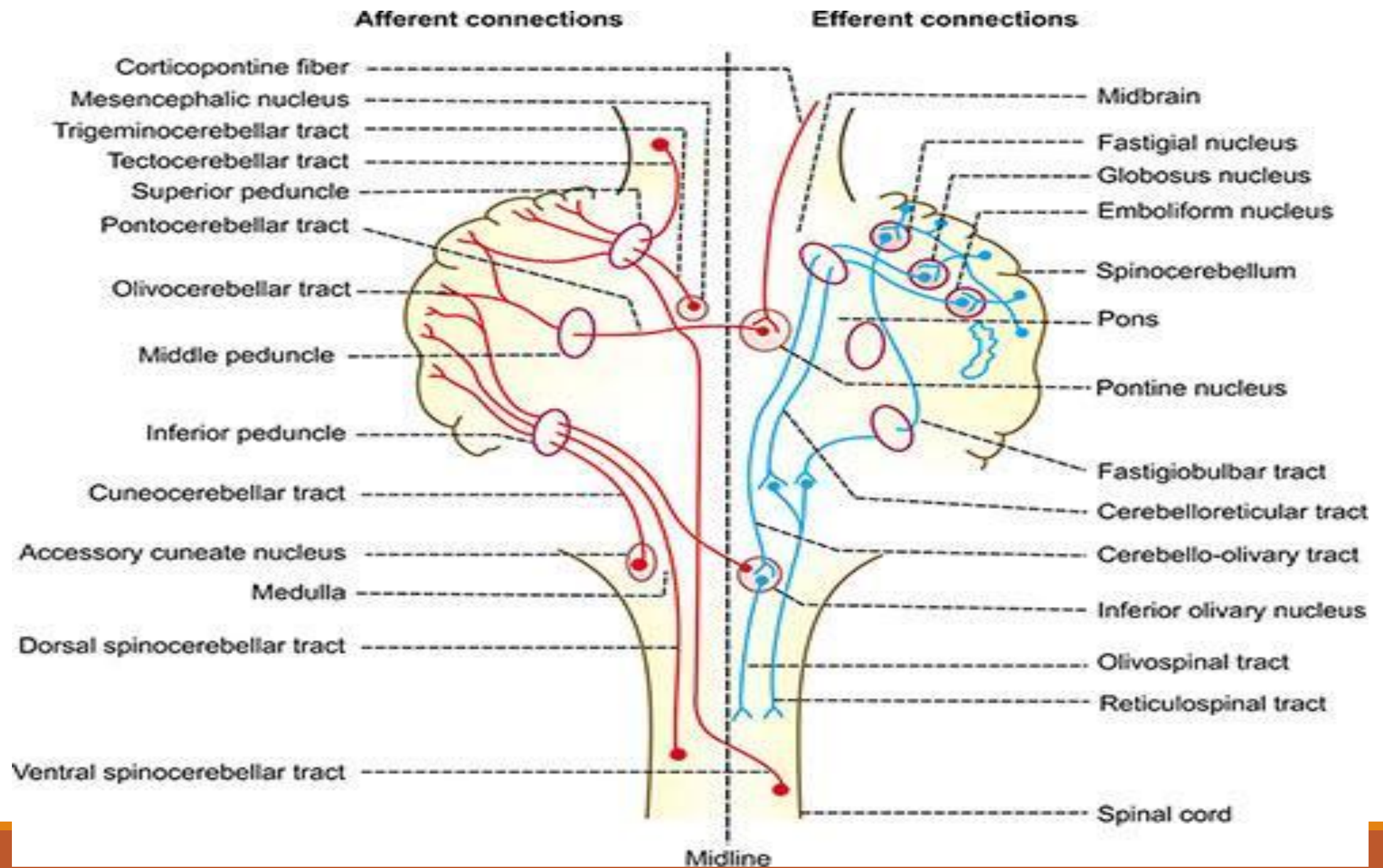
- Fibers arise from pontine nuclei, crosses the midline & reaches spinocerebellum through middle cerebellar peduncle of opposite side.
- Pontine nuclei receive afferent impulses cerebral cortex.
- Pontocerebellar tract sends information to spinocerebellum regarding motor signals discharged from cerebral cortex.

Tectocerebellar tract

- Fibers arise from superior & inferior colliculi of tectum in midbrain & reaches spinocerebellum through superior cerebellar peduncle of same side.
- Tectocerebellar tract conveys visual impulses from superior colliculus & auditory impulses from inferior colliculus to spinocerebellum.

Trigeminocerebellar tract

- Fibers of this tract originate from mesencephalic nucleus of trigeminal nerve & reaches spinocerebellum through superior cerebellar peduncle of same side.
- This tract transmits proprioceptive signals from jaw muscles & TMJ to spinocerebellum & also the sensory impulses from periodontal tissues to spinocerebellum.



EFFERENT CONNECTIONS

Fastigiobulbar tract

- Fibers arise from fastigial nucleus & passes through inferior cerebellar peduncle of same side to reach reticular formation.

Cerebelloreticular tract

- Fibers arise from nucleus emboliformis & nucleus globosus, pass through superior cerebellar peduncle of same side & end in reticular formation.
- From reticular formation, reticulospinal tract arise & end on gamma motor neurons of spinal cord.

Cerebello-olivary tract

- Fibers arise from nucleus emboliformis & nucleus globosus to reach the inferior olivary nucleus of same side through superior cerebellar peduncle.
- Olivospinal tract arises from olivary nucleus & fibers of this tract terminate on alpha motor neurons of spinal cord.

FUNCTIONS OF SPINOCEREBELLUM

- ❑ Spinocerebellum maintains tone, posture & equilibrium by receiving sensory impulses from tactile receptors, proprioceptors, visual receptors & auditory receptors.
- ❑ Spinocerebellum is the receiving area for tactile, proprioceptive, auditory & visual impulses & also receives cortical impulses through pontine nuclei.
- ❑ Spinocerebellum controls postural reflexes by modifying muscle tone & facilitates discharge from gamma motor neurons of spinal cord.
- ❑ Increased discharge from gamma motor neurons increases muscle tone whereas lesion, destruction or abolishing the function of spinocerebellum results in hypotonia & postural disturbances.
- ❑ As spinocerebellum also receives impulses from optic & auditory pathway, it helps in adjustment of posture & equilibrium in response to visual & auditory impulses.

CORTICOCEREBELLUM (NEOCEREBELLUM)

- It is the largest part of cerebellum.
- It is called ***corticocerebellum*** or ***cerebrocerebellum*** due to its connection with cerebral cortex.
- It includes lateral portions of cerebellar hemispheres.
- It is associated with planning, programing & coordination of skilled movements.

CONNECTIONS OF NEOCEREBELLUM

AFFERENT CONNECTIONS

Pontocerebellar tract

- Largest tract in the body & fibers arise from pontine nuclei, cross the midline & reach corticocerebellum through middle cerebellar peduncle off opposite side.
- It receives signals from motor area of cerebral cortex & transmits them to corticocerebellum, hence called as ***corticopontocerebellar circuit***.

Olivocerebellar tract

- Fibers arise from inferior olivary nucleus in the medulla, crosses midline to reach corticocerebellum via inferior cerebellar peduncle of opposite side & terminate on dentate nucleus & cerebellar cortex.
- Inferior olivary nucleus receives impulses from brainstem, spinal cord & cerebral cortex & transmit these impulses to corticocerebellum via olivocerebellar tract.
- This tract is formed by ***climbing fibers***.

EFFERENT CONNECTIONS

- Efferent signals from corticocerebellum are conveyed mainly via ***dentate nucleus***.
- Fibers from dentate nucleus pass via superior cerebellar peduncle, cross the midline to form decussation with the fibers of opposite side & then divide into two tracts:

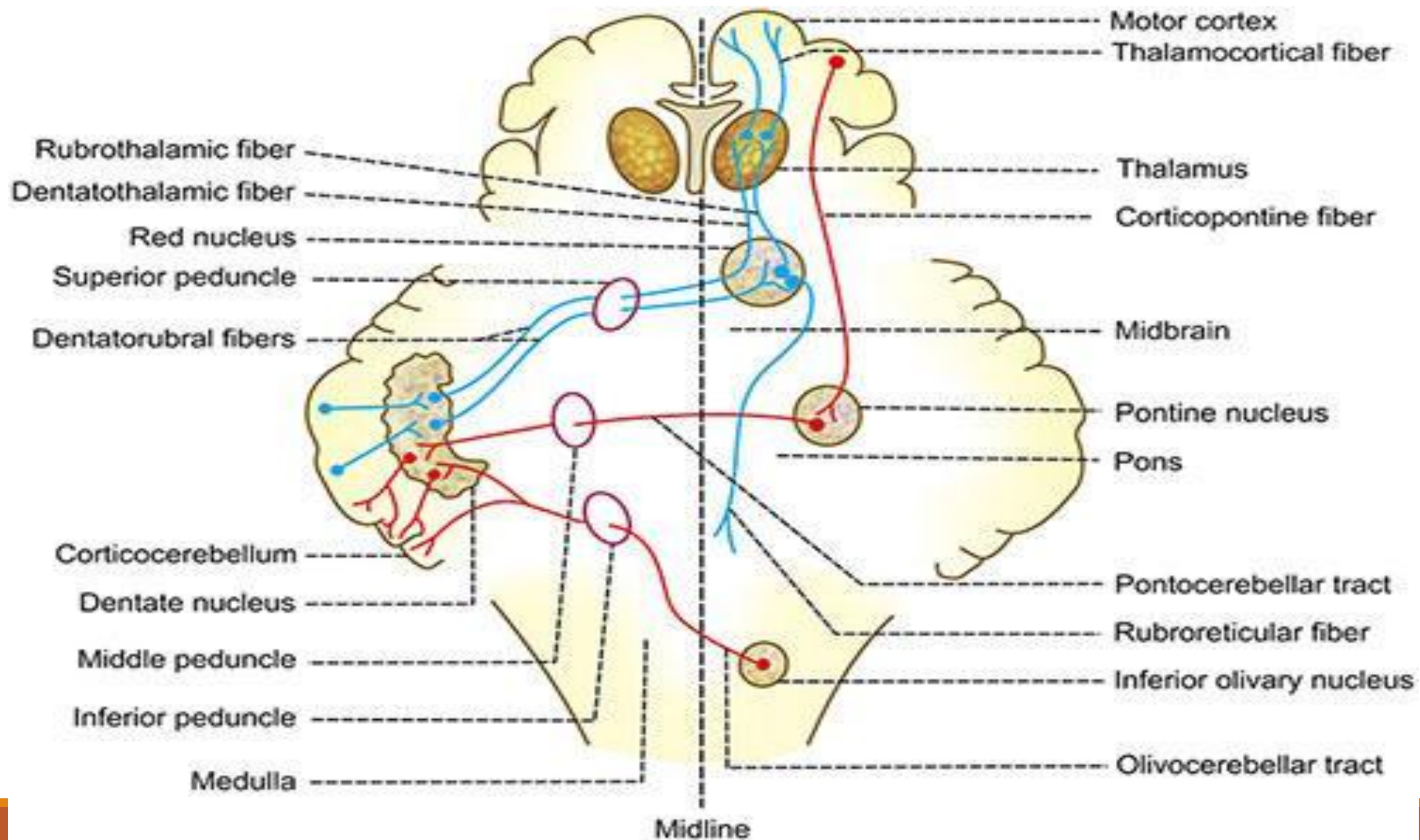
Dentatothalamic tract

- In this tract, fibers pass through ***red nucleus*** without synapsing & terminate in lateral ventral nucleus of thalamus.
- From thalamus, efferent signals then pass through ***thalamocortical fibers*** into motor area of cerebral cortex.

Dentatorubral tract

□ After crossing the midline, fibers terminate in the red nucleus as dentatorubral tract & then project via three tracts:

- I. **Rubrothalamic tract:** originate from red nucleus & terminate in lateral ventral nucleus of thalamus to project into the cerebral cortex via thalamocortical fibers.
- II. **Rubroreticular tract:** descends down to end in reticular formation & projects further into spinal cord via reticulospinal tract.
- III. **Rubrospinal tract:** projects directly into the spinal cord from red nucleus.

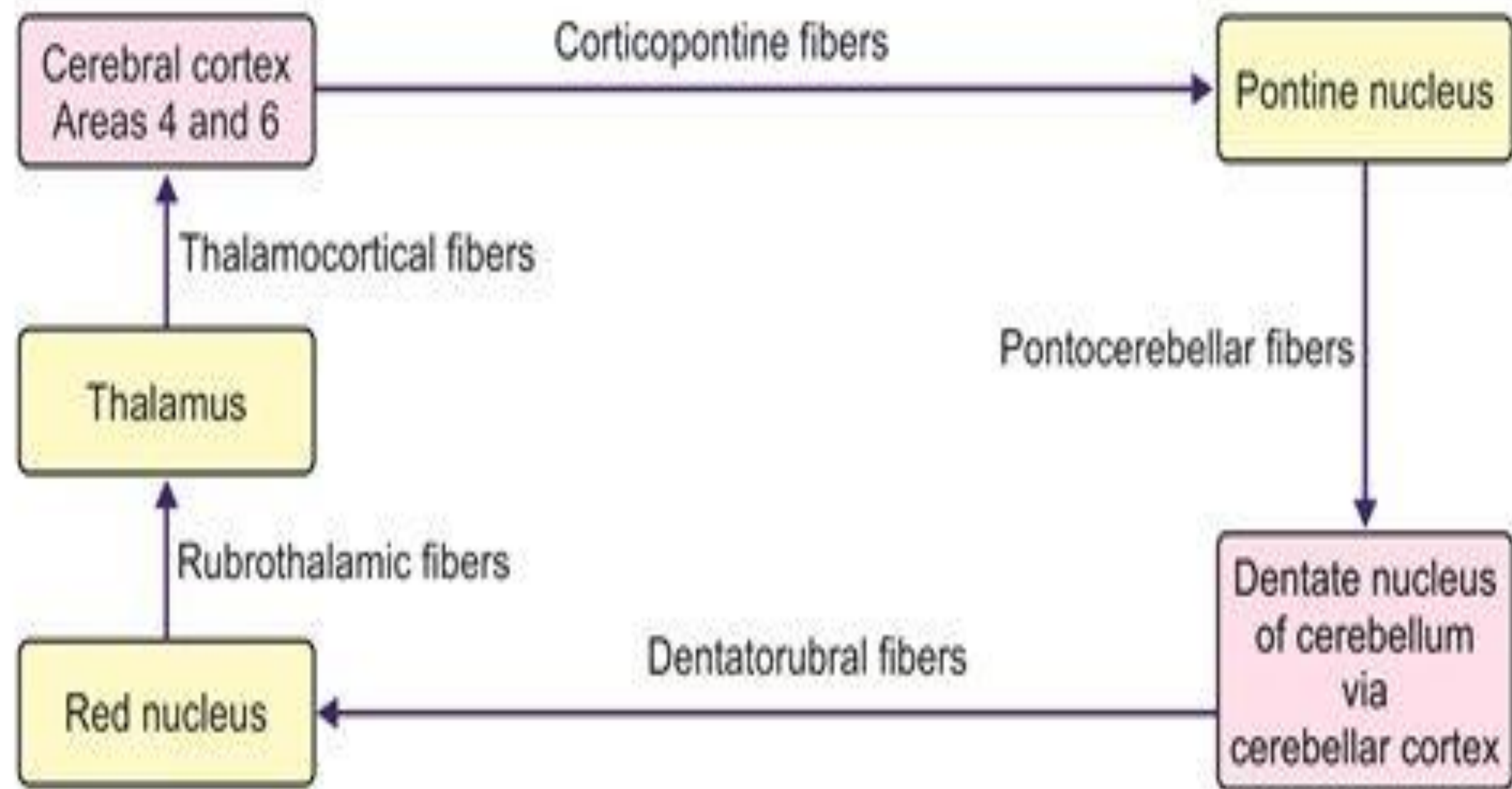


AFFERENT-EFFERENT CIRCUIT

- Afferent-efferent circuit also called ***cerebro-cerebello-cerebral connection*** is an important neuronal pathway involved in cerebellar control of voluntary movements initiated by motor area of cerebral cortex.
- This circuit includes two tracts, namely

Cerebropontocerebellar tract: fibers from motor areas 4 & 6 in frontal lobe of cerebral cortex enter the pontine nuclei & pass via middle cerebellar peduncle of opposite side as pontocerebellar fibers to end in the cerebellar cortex.

Dentatorubrothalamocortical tract: fibers from dentate nucleus pass through superior cerebellar peduncle to end in red nucleus of opposite side as dentatorubral fibers & project into thalamus as rubrothalamic fibers which is connected to areas 4 & 6 in motor cortex of cerebrum via thalamocortical fibers.



FUNCTIONS OF CORTICOCEREBELLUM

Damping action

- Corticocerebellum receives impulses from motor cortex as well as feedback signals from muscles to prevent exaggerated muscular activity by cutting-off any extra impulses.

Control of ballistic movements

- Rapid alternate movements that take place in different parts of body while doing any skilled work like typing, cycling, dancing etc are called ballistic movements.

Timing & programming of movements

- While doing any fast-skilled work(e.g. using a typewriter)a chain of movements occur rapidly in a sequential manner.
- Corticocerebellum plans these sequential movements by planning schedule of time duration of each movement & time interval between movements.

Cont..

Servomechanism

- Servomechanism is the correction of any disturbance or interference while doing skilled work.

Comparator function

- Cerebellum receives information from cerebral cortex regarding cortical impulses sent to muscles as well as receives proprioceptive impulses from muscles regarding their actions.
- By receiving signals from both ends, corticocerebellum compares the cortical commands for muscle activity & the actual movements performed by muscles.
- If any correction is to be done then corticocerebellum send impulses to the motor cortex to correct or modify the commands to muscles.

Motor learning

- Cerebellum helps the body practice & remember complex skills (like riding bike or typing etc) so they become automatic over time.

CEREBELLAR LESIONS

DISTURBANCES IN TONE & POSTURE

Atonia or hypotonia

- Loss or reduction in muscle tone due to cerebellar lesion causes disturbances in postural reflexes.

Changes in attitude

- Rotation of head towards the unaffected side.
- Lowering of shoulder & abduction(outward movement)of leg on the affected side.
- Weight of body is placed on leg of unaffected side resulting in trunk bent with concavity towards the affected side.

Cont..

Lateral deviation of arms

- In unilateral lesion, arm of affected side deviates laterally while in bilateral lesion, both arms deviate laterally when both arms are stretched & held in front of body with closed eyes.

Effect on deep reflexes

- Pendular movements occur while eliciting a tendon jerk (especially knee jerk, patellar tendon reflex).

DISTURBANCES IN EQUILIBRIUM

- Legs are spread to provide a broad base while standing & body sways side to side with oscillations of head.
- A staggering, reeling (drunken like) walking pattern is observed in cerebellar lesion.

DISTURBANCES IN MOVEMENTS

Ataxia: lack of coordination of movements.

Asynergia: lack of coordination between different muscle groups i.e. protagonists, antagonists, synergists.

Asthenia: weakness, fatigued easily & slowness of muscles.

Hypermetria/hypometria: the arm may overshoot or it may fall short while reaching an object.

Intention tremor: tremor while attempting any voluntary action.

Astasia: unsteady voluntary movements.

Nystagmus: to & fro movement of eyeball.

Rebound phenomenon: when patient attempts to do movement against resistance, the limb moves forcibly in the direction of attempt if resistance is removed suddenly.

Dysarthria: disturbances in speech.

Adiadochokinesis: inability to do rapid alternate successive movements.

POST ASSESSMENT

Q1. The deep cerebellar nucleus associated with vestibulocerebellum is:

- a) Fastigial
- b) Emboliform
- c) Globose
- d) Dentate

Q2. Damage to cerebellum most commonly produce:

- a) Paralysis
- b) Ataxia
- c) Aphasia
- d) hemianopia

Cont..

Q3. The spinocerebellar tract mainly conveys information about:

- a) Vision
- b) Hearing
- c) Taste
- d) Proprioception

Q4. Corticopontocerebellar fibers enter the cerebellum through:

- a) Superior cerebellar peduncle
- b) Middle cerebellar peduncle
- c) Inferior cerebellar peduncle
- d) Medial longitudinal fasciculus